POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Spring

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Buffer solution with types and mechanism of working. 7

   b) Define standard electrode potential. The standard reduction potential of Cu^{++}/Cu and Ni^{++}/Ni electrodes are +0.34V and -0.25V respectively. Constuct a galvanic cell using these electrodes. Write the cell reaction and calculate the standard emf of the cell. For what concentration of Cu^{++}, the emf will be zero at 25°C if the concentration of the Ni^{++} is 0.01M. 8

2. a) Differentiate between ionization energy and electron affinity. Why and in what ways does lithium resembel magnesium. 8

   b) Give reasons: 7

      i. Zinc is not consider as a true transition element.

      ii. $[Mn(OH)_6]^{2+}$ is pale pink, MnO_2 is black and MNO_4^- is intensely purple colour.

3. a) Differentiate carbocation and carbanion in terms of formation, structure and stability. 7

   b) Show the detail mechanism and role of solvent of SN1 reaction and SN2 reaction. 8

4. a) Write notes on Bakalite, Nylon and polyurethelene. 7

   b) Describe the method of processing of Natural rubber and vulcanization. 8

5. a) Describe the principle of mass spectroscopy. Give reason why ^{12}C carbon doesn't show NMR phenomenon while ^{13}C shows NMR phenomenon though the relative abundances of ^{12}C and ^{13}C carbon are 98.9% and 1.1% respectively. 8

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b) Define lubricants and cements with examples. Describe how cement is manufactured in industry. 7

6. a) Define Ozone layer with its importance. Describe how does it formes and depeletes.

b) Show your aquitance with cause of water pollution and its effect in human health. What are the control measures of water pollution. 8

7. Write short notes on: **(Any two)** 2×5

a) Thin Layer chromatography

b) Polymerization

c) Optical activity